

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

SATURDAY, 14 MAY 2022

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHOLOER OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) questions in section B, marks will only be awarded for the FIRST questions in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**SECTION A**

Q1. (a) In a C++ program, variables are declared as below:

```
int a = 10, b = 2, c = 4;
double d = 9.6, e = 2.5;
```

With the declared variables,

(i) Define the outputs of the following codes. (4 marks)

```
cout << a + b << endl;
cout << d - e << endl;
cout << c * d << endl;
```

(ii) Define the library that you need to include in order to produce the values of (3) and (4) as shown in the output segment below.

Welcome to Division Section (1) 10 / 2 = 5 (2) 10 / 4 = 2 (3) 4.00 / 2.50 = 1.60 (4) 2.00 / 9.60 = 0.208

(2 marks)

(iii) With the declared variables, write the codes that able to COUT the exact outputs as shown in the segment above using the proper stream manipulators. (9 marks)

(b) Write a complete C++ program (without using array and function) that allow a teacher to input marks of five subjects and calculates its average as shown below:

Please key marks of five subjects Subject 1: 88 Subject 2: 65.5 Subject 3: 97 Subject 4: 40 Subject 5: 67 Average mark = 71.50

(10 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

Q2. (a) Given a variable **x = 88**, evaluate the final value of the following expressions:

(i) `x % 2 + 3` (2 marks)

(ii) `12 + x * 5 / float(4)` (2 marks)

(iii) `((x - 6) % 10 / 5.0)` (2 marks)

(iv) `static_cast<int>(x * 4.3 / 7)` (2 marks)

(b) Given that:

```
int z = 15, y = 6;
z /= --y;
```

Show step-by-step evaluations of the following expressions. Treat each of the **if** statement individually.

(i) `if (z-- % 6 <= y && true)` (4 marks)

(ii) `if (!(--y < 4 || !(2 == z++)))` (4 marks)

(c) Write a complete C++ menu-driven program which allows users to select their choice of the set lunch meal from the list as shown in Table 2.1. They should key in the number of sets that they want to purchase and also select the type of free drink. Users are allowed to re-enter the choice if their input is not valid. A sample output is shown in Figure 2.1.

Table 2.1. Set Lunch Menu

Set Lunch	Price (RM)
Japanese Chicken Curry	29.90
Oyakodon (Chicken and Egg Bowl)	24.50
Cheese Katsu with Rice	26.80
Kimchi Fried Rice with Ebi Fry	33.50

**with free coffee or iced tea

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q2. (c) (Continued)**

```
Set Lunch Menu:
1. Japanese Chicken Curry
2. Oyakodon (Chicken and Egg Bowl)
3. Cheese Katsu with Rice
4. Kimchi Fried Rice with Ebi Fry
Pick a choice from the list: 1
Number of set(s): 2
Please choose coffee or iced tea (1-coffee 2-iced tea): 1
Please choose coffee or iced tea (1-coffee 2-iced tea): 2
Do you want to select another set meal? (Y-Yes, N-No)
Y
Pick a choice from the list: 5
Invalid choice! Please try again later.
Pick a choice from the list: 3
Number of set(s): 1
Please choose coffee or iced tea (1-coffee 2-iced tea): 2
Do you want to select another set meal? (Y-Yes, N-No)
N
Please pay RM 86.60
```

(Hint: Y = Yes; N=No)

Figure 2.1. Sample Output

(9 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q3. A table which describes the status of Malaysians in few states who received COVID-19 vaccine (since February 2021 until January 2022 by “covidnow.moh.gov.my/vaccinations/”) is being displayed in Table 3.1.

Table 3.1: Status of vaccinations for Klang Valley, Penang, Perak, Johor and Sarawak. (Kuala Lumpur, Selangor and Putrajaya are considered as one state under Klang Valley in this case)

	State	Population of the State (%) (Over Country Population)	At Least Two Doses (%) (Over State Population)
1.	Klang Valley	25.82	94.50
2.	Penang	5.42	86.00
3.	Perak	7.69	74.10
4.	Johor	11.61	80.70
5.	Sarawak	8.64	75.80

*Assume the population of Malaysia is 32.65 million and 1 million is 1,000,000.

Assuming a program related to the description above is implemented and the **MAIN** function is shown as below:

```
#define states 5 //size of array (defining the number of states)

int main() {

    //(A) Declare the array to store and initialize only with the values as shown in
    Table 3.1: 1) array of the population of each state, 2) array of at least
    two doses of each state, and 3) array of the name of each state.

    //(B) Declare the variables: 1) array to store the population that have not
    received at least two doses of vaccine of each state (%), and 2) the
    population of Malaysia.

    //(C) Function call sumForTwoDoses(pOfState, pOfStateWithTwoDoses,
    nameOfState, pMsia);

    //(D) Function call withoutTwoVaccine(pOfState, pOfStateWithTwoDoses,
    pWithoutTwoVaccine, nameOfState);

    //(E) Function call rankingWithoutTwoVaccine(pWithoutTwoVaccine,
    nameOfState);

    return 0;
}
```

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q3. (Continued)**

The following are the descriptions for A, B, C, D and E sections in **MAIN** function including the **THREE (3)** functions used.

- A. Declare the array to store and initialise only with the values as shown in Table 3.1:
 1) array *pOfState* to store the percentage of population of each state, 2) array *pOfStateWithTwoDoses* to store the percentage of at least two doses of each state, and 3) array *nameOfState* to store the name of each state.
NOTE: initialise in the order as shown in the table.
- B. Declare the variables: 1) array *pWithoutTwoVaccine* to store the percentage of population which have not received two doses of vaccine of each state, and 2) *pMsia* which define the population of Malaysia.
- C. ***void sumForTwoDoses(double pOfState[], double pOfStateWithTwoDoses[], string nameOfState[], int pMsia)***
 Calculate and display (in figure) the total population that have received at least two doses for all the states listed in Table 3.1.
- D. ***void withoutTwoVaccine(double pOfState[], double pOfStateWithTwoDoses[], double pWithoutTwoVaccine[], string nameOfState[])***
 Compute the percentage (%) of the population of each state which have not received at least two doses of vaccine (over country population) and store in the array *pWithoutTwoVaccine*. The details are to be displayed.
- E. ***void rankingWithoutTwoVaccine(double pWithoutTwoVaccine[], string nameOfState[])***
 Comparison is carried out using *pWithoutTwoVaccine* which obtain from function *withoutTwoVaccine* in order to arrange the ranking of the five states in descending order. The details are saved into a text file named "Without2Doses.txt".

Based on the definition above, answer the following questions by providing C++ codes.

- (a) Declare all the variables as described in the **section A** and **section B**. (8 marks)
- (b) Write function definition for *sumForTwoDoses* to calculate and display the total population that received at least two doses for all the listed states as in Figure 3.1.

Total population of the five states with at least two doses: 16546292

Figure 3.1: Output of the total population of the five states which have received at least two doses of vaccine.

(5 marks)

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICESQ3. (Continued)

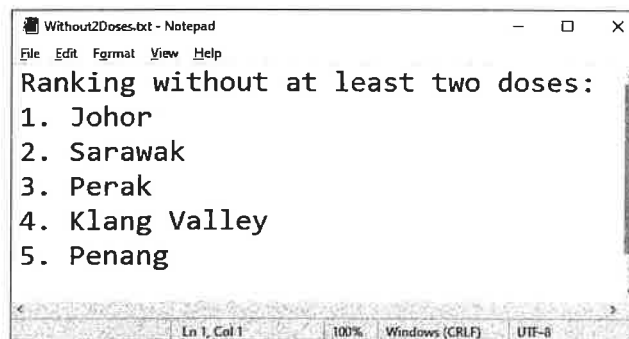
- (c) Write function definition for *withoutTwoVaccine* to compute and display the percentage of population which have not received at least two doses of vaccine as in Figure 3.2.

```
Population without at least two doses (over country population):
Klang Valley --> 1.42 %
Penang --> 0.76 %
Perak --> 1.99 %
Johor --> 2.24 %
Sarawak --> 2.09 %
```

Figure 3.2: Output of the percentage of population over the whole Malaysia for each state which have not received at least two doses of vaccine

(4 marks)

- (d) Write function definition for *rankingWithoutTwoVaccine* to compare and save the ranking in descending order as shown in Figure 3.3.



```
Without2Doses.txt - Notepad
File Edit Format View Help
Ranking without at least two doses:
1. Johor
2. Sarawak
3. Perak
4. Klang Valley
5. Penang
Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Figure 3.3: Text file named “Without2Doses.txt” which save the ranking of the states which have not received at least two doses of vaccine.

(8 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**SECTION B**

- Q4. (a) Define a struct called ***FreshFood*** that consists of three variables: *type* as ***string***, *place* as ***character array*** and *price* as ***float***. (5 marks)
- (b) In C++, declare an array of ***FreshFood*** type called *food* that is defined in Q4(a) with 50 elements. (3 marks)
- (c) Based on the struct ***FreshFood*** defined in Q4(a), write a function named ***void writeFreshFoodDetails(FreshFood food[], int &noFood)*** that stores the information of user's input into the struct array declared in Q4(b). Then, write the information of fresh food into a text file named *foodDetails.txt* as shown in Figure 4.1.

```

YELLOW ONIONS
HOLLAND
4.00
RED ONION
INDIA
4.00
RED / SMALL ONION
INDIA
4.00
RED / SMALL ONION
THAI
10.00
GARLIC
CHINA
7.00

```

Figure 4.1. foodDetails.txt

The information in Figure 4.1 is arranged using the repeated structure as shown in Figure 4.2.

```

Type of Fresh Food
Production Place
Price per kg

```

Figure 4.2. Structure of the content in foodDetails.txt

(7 marks)

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q4. (Continued)**

- (d) Write a function named ***void findFreshFoodDetails(FreshFood food[], int noFood)*** that prompt the user to input the fresh food details that he/she wants to view the price. Display the fresh food, production place and the total price of the selected fresh food according to kilogram as shown in Figure 4.3.

```
Please enter the type of fresh food you wish to purchase: RED
ONION
Please enter the production place of RED ONION: INDIA
Please enter the weight in kg that you want to view the price
of the fresh food :3.5
Fresh Food: RED ONION
Production Place: INDIA
Total price of 3.5kg is RM 14.00
```

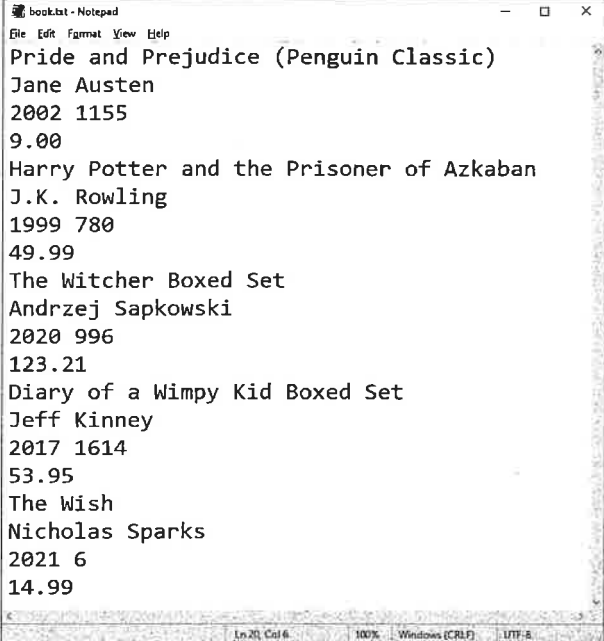
Figure 4.3. foodDetails.txt

(10 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q5. (a) Using C++ define a struct called **bookStore** that consists of 5 types of variables; **title of the book** as string, **author of the book** as string, **year of publication** as integer, **best seller rank** as integer and **price of the book** as double. (6 marks)
- (b) Define a function with its header **void readBook(bookStore book[], int &noBook)** that reads the text file (book.txt) as shown below and store the information into the **bookStore** using struct array:

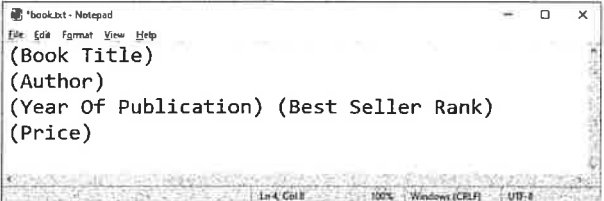


```

book.txt - Notepad
File Edit Format View Help
Pride and Prejudice (Penguin Classic)
Jane Austen
2002 1155
9.00
Harry Potter and the Prisoner of Azkaban
J.K. Rowling
1999 780
49.99
The Witcher Boxed Set
Andrzej Sapkowski
2020 996
123.21
Diary of a Wimpy Kid Boxed Set
Jeff Kinney
2017 1614
53.95
The Wish
Nicholas Sparks
2021 6
14.99
Ln 20, Col 6      100% Windows (CRLF) UTF-8

```

The information is arranged based on the following structure:



```

*book.txt - Notepad
File Edit Format View Help
(Book Title)
(Author)
(Year Of Publication) (Best Seller Rank)
(Price)
Ln 4, Col 8      100% Windows (CRLF) UTF-8

```

(5 marks)

- (c) Define a function with its header **void displayBook(bookStore book)** that able to display the details of a book as shown below:

Title: The Wish Author: Nicholas Sparks Year of Publication: 2021 Rank: 6 Price: 14.99
--

(5 marks)

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q5. (Continued)**

- (d) Define a function with its header **void bestSellerRank(bookStore book[], int noBook)** that determine the best seller from the list of books and display the details by calling **displayBook** as shown below:

```
The best seller in the list is
Title: The Wish
Author: Nicholas Sparks
Year of Publication: 2021
Rank: 6
Price: 14.99
```

(3 marks)

- (e) Define a function with its header **void selectBook(bookStore book[], int noBook)** that able to filter the list of the books by allowing the user to provide the range of the years and display the details by calling **displayBook** as shown below:

```
Please key in the range of the year
Start: 2015
End: 2020

Title: The Witcher Boxed Set
Author: Andrzej Sapkowski
Year of Publication: 2020
Rank: 996
Price: 123.21

Title: Diary of a Wimpy Kid Boxed Set
Author: Jeff Kinney
Year of Publication: 2017
Rank: 1614
Price: 53.95
```

A message will be displayed if no book is available within the range as shown below:

```
Please key in the range of the year
Start: 2012
End: 2013

Sorry, no book is available.
```

(6 marks)

[Total : 25 marks]

